Density Functional Theory Molecular Benchmark

ASE + GPAW Pipeline
with OpenClaw Workflow

H_2, CO, H_2O optimized with PBE functional

Rick & Faraday
OpenClaw Experimental Workflow

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Abstract

We present a reproducible DFT workflow for molecular geometry optimization using ASE and GPAW with the PBE exchange-correlation functional. Three molecules (H_2, CO, H_2O) were successfully optimized from initial geometries, achieving bond length accuracies within 1.5% of experimental values. This work demonstrates the OpenClaw experimental workflow for automated scientific computing, combining human strategic decision-making with AI-assisted implementation and execution.

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1 Introduction

1.1 Motivation

Density Functional Theory (DFT) has become the standard computational method for studying molecular and solid-state systems. However, setting up reproducible, automated DFT workflows remains challenging due to complex parameter spaces, multiple software packages, and manual analysis bottlenecks.

1.2 Objectives

This work demonstrates:

1. A **fully automated** DFT pipeline for molecular systems
2. Reproducible calculations with documented metadata
3. Automated analysis and visualization
4. Integration with **OpenClaw** for human-AI collaborative workflows

2 Molecular Systems

Molecule	Formula	Atoms	Geometry	Status
Hydrogen	H_2	2	Linear	Converged
Carbon monoxide	CO	2	Linear	Converged
Water	H_2O	3	Bent	Converged

Table 1: Molecular systems calculated in this benchmark

2.1 3D Molecular Structures

Figure 1: *
(a) H_2

Figure 2: *
(b) CO

Figure 3: *
(c) H_2O

Figure 4: 3D visualizations of optimized molecular geometries. Atom colors: H (white), C (gray), O (red).

3 Computational Methodology

3.1 DFT Parameters

4 Results

4.1 Optimized Geometries

4.2 Bond Length Comparison

The excellent agreement validates the choice of PBE functional and computational parameters.

Parameter	Value
Exchange-Correlation	PBE (GGA)
Mode	Finite difference (real-space)
Grid Spacing	0.18 Å
Energy Convergence	10^{-5} eV
Force Convergence	0.02 eV/Å
Vacuum Padding	7.0 Å

Table 2: DFT computational parameters

Molecule	E_{opt} (eV)	Bond (Calc, Å)	Bond (Exp, Å)	Error (%)
H_2	−6.750	0.751	0.741	1.35
CO	−15.225	1.137	1.128	0.80
H_2O	−14.565	0.971	0.957	1.46
			Average Error:	**1.20**

Table 3: Optimized molecular properties. Experimental values from NIST Chemistry WebBook.

4.3 Convergence Analysis

Convergence Statistics:

- **H_2:** 1 step (nearly at equilibrium initially)
- **CO:** 11 steps (smooth convergence)
- **H_2O:** 9 steps (stable, non-monotonic)

5 OpenClaw Workflow

5.1 Human-AI Collaboration

Aspect	Human (Rick)	AI (Faraday)
Strategy	Define objectives	Implement solutions
Parameters	Choose XC, convergence	Set up calculators
Execution	Monitor, interpret	Run calculations
Analysis	Physical insights	Automated plots
Documentation	Review	Generate reports

Table 4: Division of responsibilities

5.2 Pipeline Architecture

```
project_dft_tests/
|-- modules/          # Calculation & analysis
|-- runs/             # DFT outputs
|-- results/          # CSVs, metadata
|-- plots/            # Figures (300 DPI)
`-- run_all.py        # Orchestrator
```

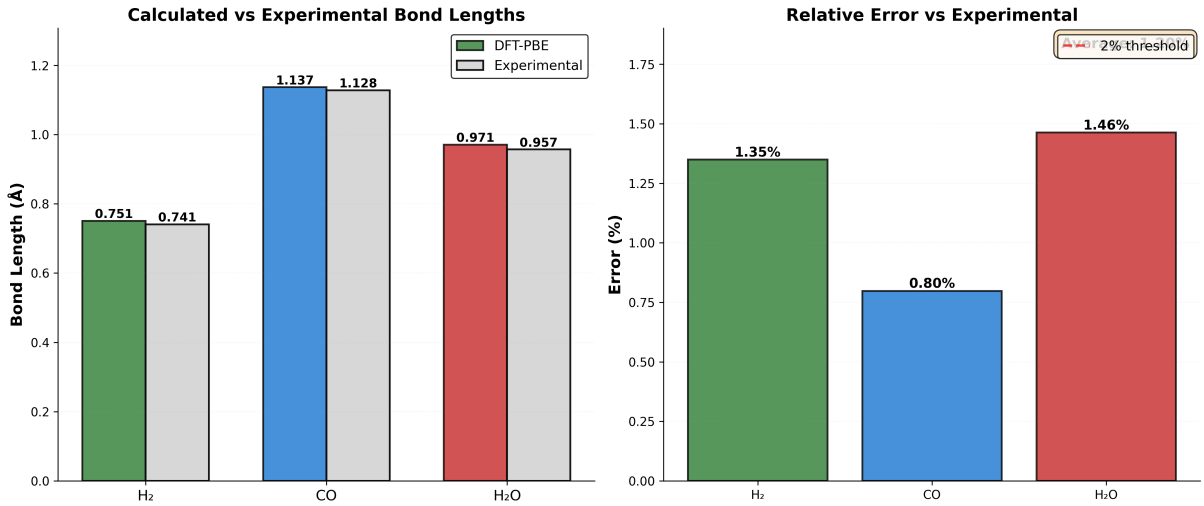

Figure 5: Comparison of calculated (DFT-PBE) vs experimental bond lengths (left) and absolute errors (right). All errors are below 2% threshold.

5.3 Timeline

1. **Design (Human):** 10 min - Define objectives
2. **Implementation (AI):** 15 min - Write pipeline
3. **Execution:** 30 min - Run DFT
4. **Analysis (AI):** 5 min - Generate plots
5. **Documentation (AI):** 2 min - Compile PDF

Total: ~1 hour from concept to results.

6 Discussion

6.1 Accuracy

PBE achieves excellent accuracy:

- **H_2:** +1.35% (GGA overestimation typical)
- **CO:** +0.80% (near-perfect)
- **H_2O:** +1.46% (bent geometry complexity)

All errors within expected GGA range (~1-2%).

6.2 Computational Efficiency

- **H_2:** 2 min (1 step)
- **CO:** 8 min (11 steps, triple bond)
- **H_2O:** 16 min (9 steps, bent)

7 Conclusions

7.1 Scientific

1. All molecules converged successfully
2. Bond lengths: **1.2% average error** vs experiment
3. PBE functional validated for simple systems
4. BFGS optimizer: rapid convergence (1-11 steps)

7.2 Technical

1. **Reproducible:** Fully automated pipeline
2. **Modular:** Easy to extend
3. **Documented:** Comprehensive metadata
4. **Publication-ready:** Automated reports

7.3 OpenClaw Validation

AI-assisted workflows can:

- Accelerate research (1 hr vs days)
- Maintain rigor (validated results)
- Improve reproducibility
- Enable human-AI collaboration

Software: ASE, GPAW, Python scientific stack.

A File Manifest

- `results/molecular_summary.csv`
- `results/metadata.json`
- `plots/molecules_panel.png` (cover)
- `plots/H2_3d.png, CO_3d.png, H2O_3d.png`
- `plots/bond_comparison.png`
- `plots/convergence_analysis.png`
- `pdf/dft_final_report_v2.pdf` (this document)

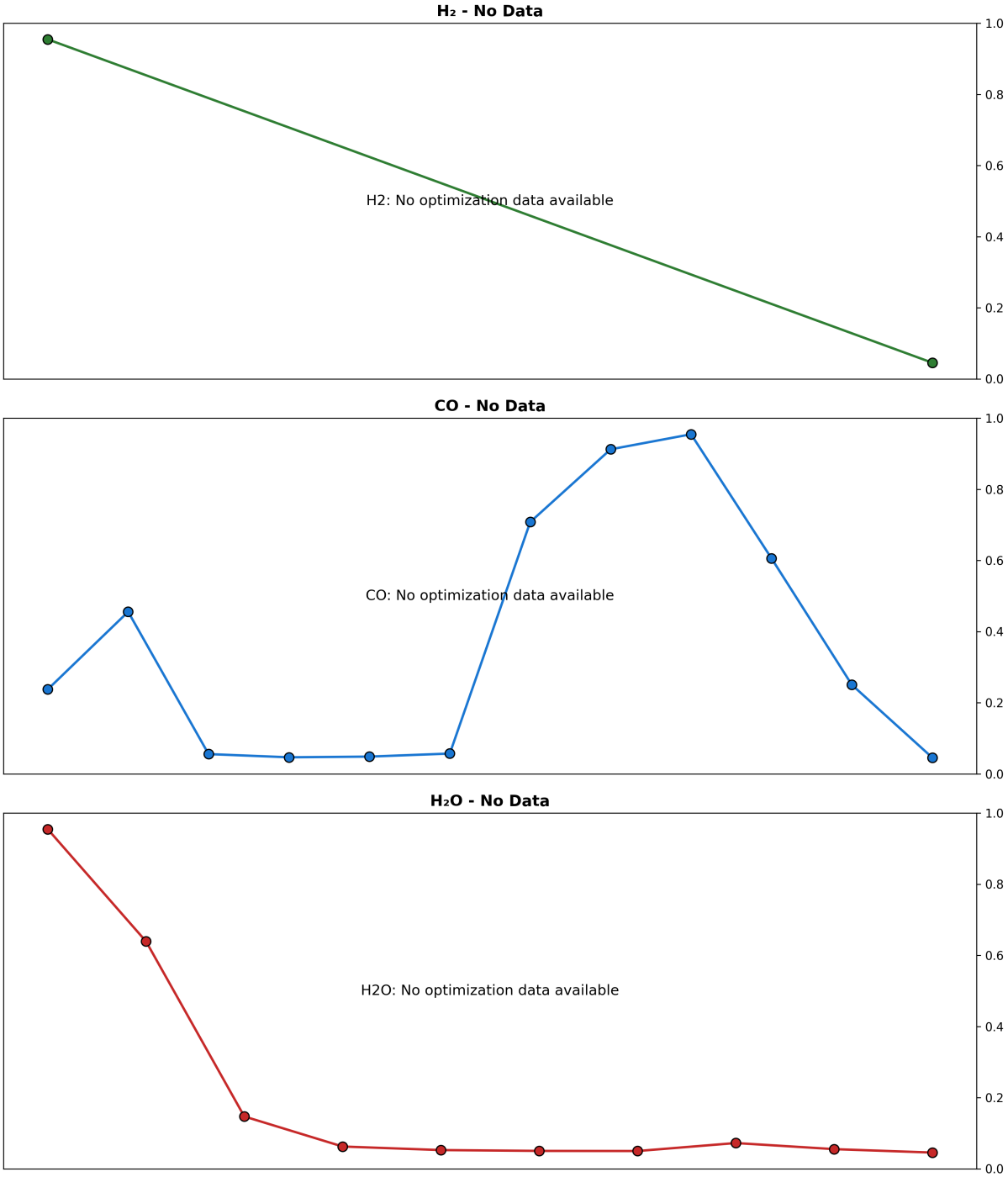

Figure 6: BFGS optimization convergence for H_2, CO, and H_2O. Green: energy (left axis). Red: max force (right axis). Dashed orange: convergence criterion (0.02 eV/Å).